

Avni Girish

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EDUCATION

Rutgers University–New Brunswick

New Brunswick, NJ

Bachelor of Science in Computer Science and Cognitive Science - Double Major

May 2026

- **Relevant Coursework:** Design and Analysis of Computer Algorithms, Data Structures and Algorithms, Discrete Structures I & II, Computer Architecture, Software Methodology, Principles of Information and Data Management, Mind/Machines & Persons, Introduction to Cognitive Science, Cognition, Intro to Artificial Intelligence, Intro to Data Science, Data Science and Data Management, Data with R
- **Organizations:** Undergraduate Alliance of Computer Science, Women in Computer Science, Rutgers University Programming Association, Women in Science and Engineering Program-Douglass Residential College

SKILLS, ACTIVITIES & INTERESTS

Languages: Python, Java, C, C#, C++, R, JavaScript, SQL

AI/ML Tools: Pandas, Hugging Face, OpenAI API, Claude 4, GPT-4, NLP, EDA, Agentic RAG, MCP, Sentiment Analysis

Frameworks: FastAPI, React.js, Next.js, Flask, Node.js, Streamlit, AngularJS

Databases: MySQL, PostgreSQL, MongoDB, SQLite, Firebase

Developer Tools: Git, Azure DevOps, MATLAB, Unity

Certifications & Training: Python Programming - *GirlsComputingLeague* (2020), First Year Fellowship - *Lead Up Academy* | *Rutgers University-New Brunswick* (2023), Software Engineering Fellowship - *Rutgers Blueprint* (2023), CS50: Introduction to Computer Science - *Harvard University* (2023), Solve Real-World Data Problems with SQL - *Linkedin* (2023), Social/Behavioral/Epidemiologic Research Investigators - *CITI Program* (2023)

Awards: 1st place in the Advanced Category - *HackSBPL* (2021)

WORK & LEADERSHIP EXPERIENCE

Food Project Research Lead

New Brunswick, NJ

Cognition and Learning Center

May 2025 - Present

- Directed a 200+ participant study, designing data protocols for cognitive assessments and microbiome analysis
- Supervised 4+ RAs in ethical data practices and data collection, reducing processing time by 40%
- Identified significant microbiome-cognition links through EDA and modeling; findings contributed to a forthcoming publication

Undergraduate Research Assistant

New Brunswick, NJ

Cognition and Learning Center

Jan 2024 - Present

- Established a 30% correlation between preschool math sense and academic math skills through computerized games and tests using **R**, **Python**, and **MATLAB**
- Identified 25% improved recall tactics in children via game-based studies through explanatory data analysis
- Revealed a 20% belief shift impacting motivation in children's cognitive development through qualitative data analysis

Rutgers Computing Academic Tutor

New Brunswick, NJ

Rutgers University Department of Computer Science

Jan 2024 - Present

- Tutored 650+ students in Data Structures, Algorithms, and **Java** through tailored strategies using visual aids and personalized environments, contributing to a 15% increase in course pass rates
- Served as a liaison between students and faculty, identifying common learning gaps and providing structured feedback to enhance curriculum delivery

InfoSec Data Science & AI Engineer Intern | AI Hub and Digital Engineering

Lake Mary, FL

The Bank of New York Mellon Corporation

Jun 2025 - Aug 2025

- Led development of AURA, a self-optimizing UI system combining **Claude 4** and **GPT-4.1**, reducing dashboard development from days to minutes
- Engineered AI-driven feedback loops where natural language inputs dynamically regenerate UI components, cutting revisions by 65% and boosting analyst productivity 3.5x
- Built an intelligent agent framework with **React.js** and **FastAPI** where AI autonomously routes tasks between planning (Claude 4) and programming (GPT-4.1) AI agents

IT Business Analyst Intern

Trenton, NJ

New Jersey Economic Development Authority

Jun 2024 - Aug 2024

- Collected and analyzed business requirements for projects exceeding \$500,000, translating them into user stories using **Azure DevOps**, **Dynamics365**, and **Visio**
- Coordinated with 5 business units and a 10-person development team, managing roadblocks and facilitating release planning for 20+ project milestones
- Conducted competitor research on state policies to inform strategic discussions impacting 250+ EDA applicants

PROJECTS

AI-Powered Personal Portfolio & Interactive Resume

Aug 2025 - Present

- Engineered a full-stack portfolio with **Next.js 15**, **TypeScript**, and **Framer Motion**, achieving 95% mobile compatibility and <2s load times
- Integrated multi-model AI chat using OpenRouter API with intelligent fallback across **GPT-4**, **Claude**, and **Gemini**, maintaining 99.8% uptime
- Architected scalable **RESTful APIs** with automated contact processing and admin analytics dashboard, supporting 1000+ concurrent users with zero downtime

MoodSync+: A comprehensive Mental and Physical Wellness App

Dec 2024 - Present

- Built an AI-powered wellness platform with **FastAPI** and **Streamlit** that provides 50+ personalized mood-based playlists and exercise routines via Spotify API
- Implemented NLP-based sentiment analysis with **Hugging Face** and **OpenAI APIs**, achieving 95% accuracy in emotion detection for journal entries
- Created interactive dashboards with **Matplotlib** and **Plotly** to visualize mood trends for 100+ users, backed by a **PostgreSQL** database